

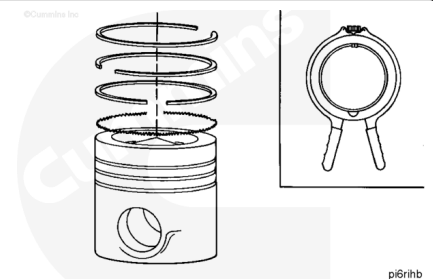
Trainings ▾

General Information

Note : Two different piston and liner combinations are available for this engine platform. The ferrous cast ductile (FCD) piston must be used in combination with a nitrided cylinder liner. The steel piston must be used in combination with a high boron liner. Each of these pistons has a corresponding piston ring set that must be used. The piston ring sets are not transferable between the different piston types.

Remove

Use piston ring expander, Part Number ST-1269, or equivalent, to remove the piston rings.



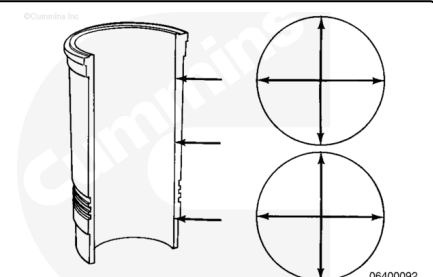
Measure

with Mechanically Actuated Injector

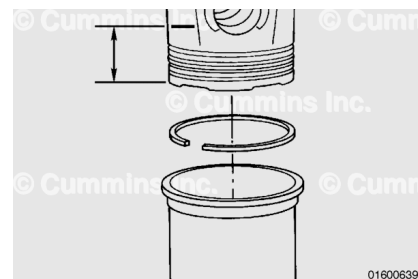
The following tasks are for engines with FCD pistons.

Verify that the liners are within specification prior to measuring piston ring gaps. Refer to Procedure 001-028 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-028-tr.html)

Note : Once a set of piston rings has been measured against a cylinder liner, they must only be used on a piston which is to be inserted into that liner. If the piston rings are subsequently required to be used in another cylinder location, the ring gap measuring process described here must be carried out again in the new cylinder liner.



Note : Engines built after engine serial number (ESN) 33175467 have a revised top piston ring and the top piston ring gap should be measured in accordance with the specification in the Measure with Electronically Actuated Injector section. Engines that have had rebuild work carried out in the past may have been upfitted with the revised top piston ring. Check the part number on the top piston ring to make sure the correct tolerances for the piston ring gap are being used.

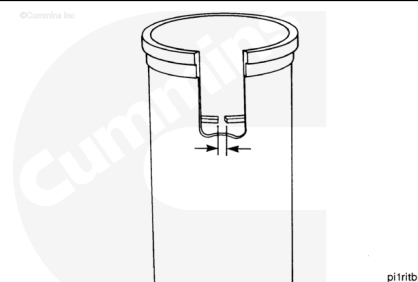


To check the ring gaps, place an ink mark on the side of a serviceable piston (with no piston rings fitted) at 100 mm [3.94 in] from the crown.

Install a piston ring in the top of a cylinder liner that meets specifications and will be used with the piston rings having their gaps measured. Use the top part of the piston to position the ring in the liner. Drop the piston down until the 100 mm [3.94 in] marking on the piston reaches the top of the liner. Carefully remove the piston and carry out the ring gap measurement.

⚠ CAUTION ⚠

If the ring gap is not within the minimum specification, do not attempt to adjust the end gap by filing or grinding. The chrome plating on the ring will be damaged and the ring will fail. Check to make sure the cylinder bore inside diameter is correct. If the end gap remains less than the minimum specification, replace the piston ring.



Use a feeler gauge to check the ring gap.

Note : Several different rings are available and used on mechanically actuated engines.

Piston Ring Gap for FCD Pistons			
	mm		in
Top 4100481	0.56	MIN	0.022
	0.76	MAX	0.03
Top 4919251	0.45	MIN	0.018
	0.65	MAX	0.026

Top 3647001	0.45	MIN	0.018
	0.6	MAX	0.024
Intermediate 3090460	0.63	MIN	0.025
	1.02	MAX	0.040
Intermediate 3645611	1.85	MIN	0.073
	2.1	MAX	0.083
Oil 3090450	0.38	MIN	0.015
	0.76	MAX	0.030

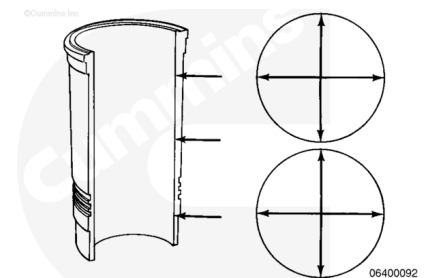
Note : Add 0.08 mm [0.003 in] to the maximum limit for each 0.03 mm [0.001 in] that the liner bore inside diameter exceeds 158.75 mm [6.25 in].

with Electronically Actuated Injector

The following tasks are for engines with FCD pistons.

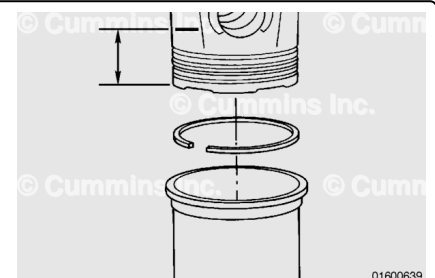
Verify that the liners are within specification prior to measuring piston ring gaps. Refer to Procedure 001-028 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-028-tr.html)

Note : Once a set of piston rings has been measured against a cylinder liner, they must only be used on a piston which is to be inserted into that liner. If the piston rings are subsequently required to be used in another cylinder location, the ring gap measuring process described here must be carried out again in the new cylinder liner.



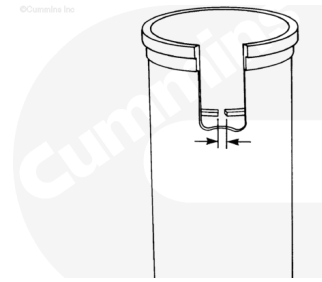
To check the ring gaps, place an ink mark on the side of a serviceable piston (with no piston rings fitted) at 100 mm [3.94 in] from the crown.

Install a piston ring in the top of a cylinder liner that meets specifications and will be used with the piston rings having their gaps measured. Use the top part of the piston to position the ring in the liner. Drop the piston down until the 100 mm [3.94 in] marking on the piston reaches the top of the liner. Carefully remove the piston and carry out the ring gap measurement.



⚠ CAUTION ⚠

If the ring gap is not within the minimum specification, do not attempt to adjust the end gap by filing or grinding. The chrome plating on the ring will be damaged and the ring will fail. Check to make sure the cylinder bore inside diameter is correct. If the end gap remains less than the minimum specification, replace the piston ring.



Use a feeler gauge to check the ring gap.

Piston Ring Gap for FCD Pistons			
	mm		in
Top	0.45	MIN	0.018
	0.65	MAX	0.026
Intermediate	0.9	MIN	0.035
	1.2	MAX	0.047
Oil	0.38	MIN	0.015
	0.76	MAX	0.030

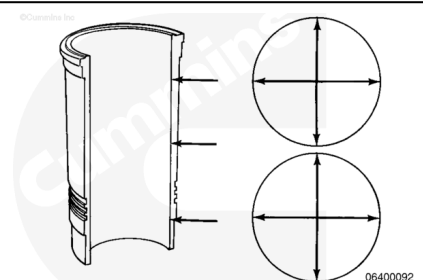
If the piston ring is **not** within specification, the piston ring **must** be replaced.

Note : Add 0.08 mm [0.003 inch] to the maximum limit for each 0.03 mm [0.001 in] that the liner bore inside diameter exceeds 158.75 mm [6.25 in].

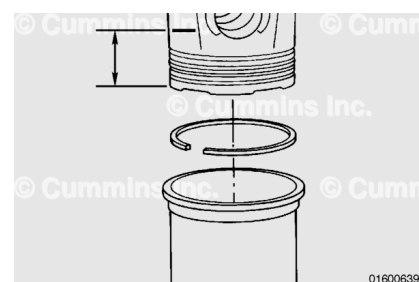
The following tasks are for engines with steel pistons.

Verify that the liners are within specification prior to measuring piston ring gaps. Refer to Procedure 001-028 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-028-tr.html>)

Note : Once a set of piston rings has been measured against a cylinder liner, they **must only** be used on a piston which is to be inserted into that liner. If the piston rings are subsequently required to be used in another cylinder location, the ring gap measuring process described here **must** be carried out again in the new cylinder liner.



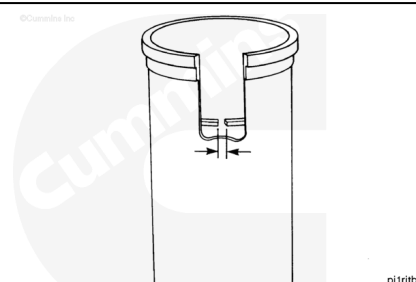
To check the ring gaps, place an ink mark on the side of a serviceable piston (with no piston rings fitted) at 100 mm [3.94 in] from the crown.



Install a piston ring in the top of a cylinder liner that meets specifications and will be used with the piston rings having their gaps measured. Use the top part of the piston to position the ring in the liner. Drop the piston down until the 100 mm [3.94 in] marking on the piston reaches the top of the liner. Carefully remove the piston and carry out the ring gap measurement.

⚠ CAUTION ⚠

If the ring gap is not within the minimum specification, do not attempt to adjust the end gap by filing or grinding. The chrome plating on the ring will be damaged and the ring will fail. Check to make sure the cylinder bore inside diameter is correct. If the end gap remains less than the minimum specification, replace the piston ring.



Use a feeler gauge to check the ring gap.

Piston Ring Gap for Steel Pistons

	mm		in
Top	0.40	MIN	0.016
	0.50	MAX	0.020
Intermediate	1.70	MIN	0.067
	1.9	MAX	0.075
Oil	0.40	MIN	0.016
	0.70	MAX	0.027

If the piston ring is **not** within specification, the piston ring **must** be replaced.

Note : Add 0.08 mm [0.003 inch] to the maximum limit for each 0.03 mm [0.001 in] that the liner bore inside diameter exceeds 158.75 mm [6.25 in].

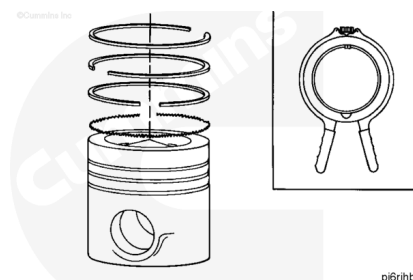
Install

Do **not** install the piston rings until the pistons are installed on the rods.

The piston rings are marked with the part number. The shipping package is marked with the location. The part number on the ring **must** be positioned toward the top of the piston.

Use a ring expander, Part Number ST-1269, or equivalent, to install the piston rings.

Note : Verify that the ring part numbers are facing toward the top of the piston on engines with steel pistons.



Rotate the rings until the gaps are positioned as illustrated.

Do **not** align the gap of a ring with the piston pin bore.

The gap in the oil ring expander **must** be turned 180 degrees opposite the gap on the oil ring.

